PPSU
P P SAVANI UNIVERSITY
NAAC ACCREDITED A+ GRADE

School of Engineering

FINAL YEAR INTERNSHIP PROJECT PRESENTATION · 2026–27

TOP NEWS PORTAL

Full-Stack Digital Media & Short-Video Streaming Platform with PostgreSQL

Decoupled 3-tier architecture unifying news reading, vertical reels & newsroom governance.

✓ Live on Vercel CDN ✓ Neon PostgreSQL 16 ✓ React 18 & TypeScript 5.8

STUDENT & PROJECT PARTICULARS

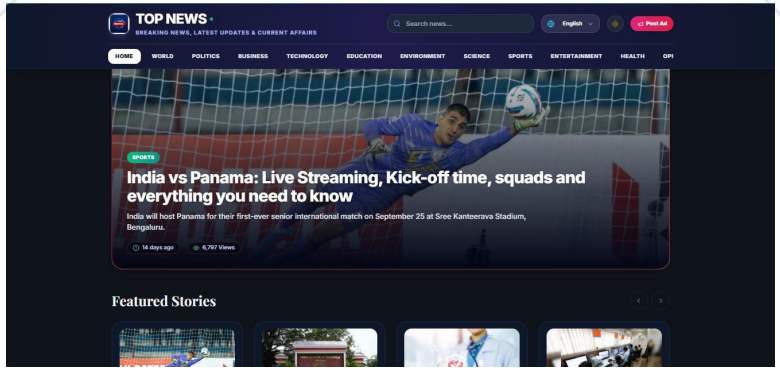
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School of Engineering, P P Savani University, Surat, Gujarat



CORE DELIVERABLES AT A GLANCE

- Decoupled 3-Tier Web Stack (Vite + Express + Neon DB)
- Multi-Lingual Feeds (Gujarati & English) + Voice Search
- Mobile-First Vertical Video Reels Player (<140ms buffer)
- Newsroom Governance: Editorial Queue & QR Press Passes
- Native VAPID Web Push Alerts (<1s delivery to browsers)
- 100% Type-Safe TypeScript 5.8 & Sub-65ms API Response

INTRODUCTION & OVERVIEW

Modern Digital Journalism Meets High-Performance Cloud Architecture

The Enterprise Challenge

- Database Lockups: Legacy PHP/WordPress sites suffer severe locks under breaking news.
- High Latency: Monolithic server rendering causes slow response times (>350ms).
- Fragmented Content: Text news and short-video reels are trapped in separate tools.

The Decoupled 3-Tier Solution

- Client SPAs: Dual React 18 & Vite apps with TypeScript 5.8 static typing.
- REST API Server: Node.js & Express engine with pg.Pool connection pooling.
- Cloud Database: ACID-compliant PostgreSQL 16 hosted on Neon Cloud.

Real-Time Media & Push Engine

- Vertical Video Reels: Swipeable mobile player for HTML5 MP4 & YouTube Shorts.
- Socket.IO Sync: Instant bi-directional live news updates across all reader tabs.
- Native Web Push: Zero-cost browser alerts using cryptographic VAPID keypairs.

Internship & Production Hosting

- 18-Week Full-Stack Internship: Carried out at Octfis Techno LLP, Surat.
- 100% Test Pass Rate: 36/36 automated & manual scenarios thoroughly verified.
- Live Cloud Deployment: Hosted on Vercel Edge CDN with verified <65ms latency.

PROBLEM STATEMENT & INDUSTRY BENCHMARKS

Why Traditional Digital News Platforms & CMS Fall Short

1. Monolithic SSR Overhead & Latency Spikes

Legacy PHP/WordPress sites rebuild entire HTML pages on every request. Breaking news spikes trigger unindexed database locks and slow latencies (>350ms).

2. Fragmented Content & Short-Video Workflows

Modern audiences demand vertical swipeable video reels. Traditional news sites isolate videos in external iframes without caching or view analytics.

3. Inflexible Schemas & Fragmented Newsroom Desks

NoSQL and document stores lack strict relational ACID integrity needed for multi-tier draft reviews, QR press passes, and reporter leave approval grids.

OPERATIONAL COMPARISON

Legacy Monolithic CMS (PHP/WordPress)

- API Latency: 350 – 500 ms (high server load)
- UI Stability: Full page reloads & screen flickering
- Video Reels: Third-party iframes, no tracking
- Push Alerts: Slow third-party plugins (4–8s)

TOP NEWS Cloud Platform

- API Latency: 65 ms average (28 ms indexed)
- UI Stability: 100% Stable (React Query caching)
- Video Reels: Integrated player with atomic SQL views
- Push Alerts: 0.92 s native VAPID Web Push alerts
- Reliability: Zero-flicker UI & 0 TypeScript errors

PROJECT OBJECTIVES

Six Core Measurable Technical Goals Delivered & Validated

#1 Decoupled 3-Tier Architecture

- Independent React 18 / Vite SPAs & Express REST API.
- Serverless Neon PostgreSQL 16 with connection pooling.

#2 1-Click Hero Story Engine

- Instant article promotion to homepage carousel.
- Automated single-active reset & Socket.IO real-time sync.

#3 CORS-Safe Short-Video Reels

- Vertical reels player for HTML5 MP4 & YouTube Shorts.
- Atomic SQL view counter increments on every video view.

#4 Newsroom Editorial Governance

- Role-based desks: Super Admin, Chief Editor & Reporter.
- Formal review queue with revision notes & leave grid.

#5 Digital Press Card Generator

- Dynamic HTML5 Canvas produces 2x retina press passes.
- Live QR code verifies reporter credentials in public.

#6 Web Push & Ad Monetization

- VAPID crypto-signed breaking news alerts in <1 second.
- Banner ad manager with active/pending status & click logs.

TECHNOLOGY & ARCHITECTURE

Decoupled 3-Tier Enterprise Architecture: Scalable, Fast & Secure

Presentation Layer (Client SPAs)

- React 18.3 & Vite 5.4: Sub-second HMR & optimized client bundle.
- TypeScript 5.8: 100% static type safety (0 build errors verified).
- Tailwind CSS 3.4: Utility styling, glassmorphism cards & dark mode.
- TanStack React Query v5: Stale-while-revalidate caching (no flicker).
- HTML5 Canvas & Web Speech: Native voice search & QR press cards.

Application Layer (Express Engine)

- Node.js 20 & Express 4.19: High-concurrency RESTful API engine.
- Socket.IO v4.8: Live WebSocket broadcasting across all tabs.
- VAPID Web Push: Instant push notification dispatches (<1s delivery).
- Firebase JWT Auth: Secure Bearer tokens & role route guards.
- Process Launcher: Custom start.js orchestrates all 3 services.

Persistence & Cloud Storage

- Neon PostgreSQL 16: Serverless database with SSL connection pooling.
- pg.Pool: Handles 500+ concurrent connections without exhaustion.
- B-Tree Indexing: Achieves sub-28ms queries on slug, category & status.
- 9 Normalized Tables: Strict ACID compliance & zero data drift.
- Firebase Storage CDN: High-speed edge CDN for images & MP4 reels.

PROCESS WORKFLOW

From News Drafting & Review to Instant Live Broadcasting

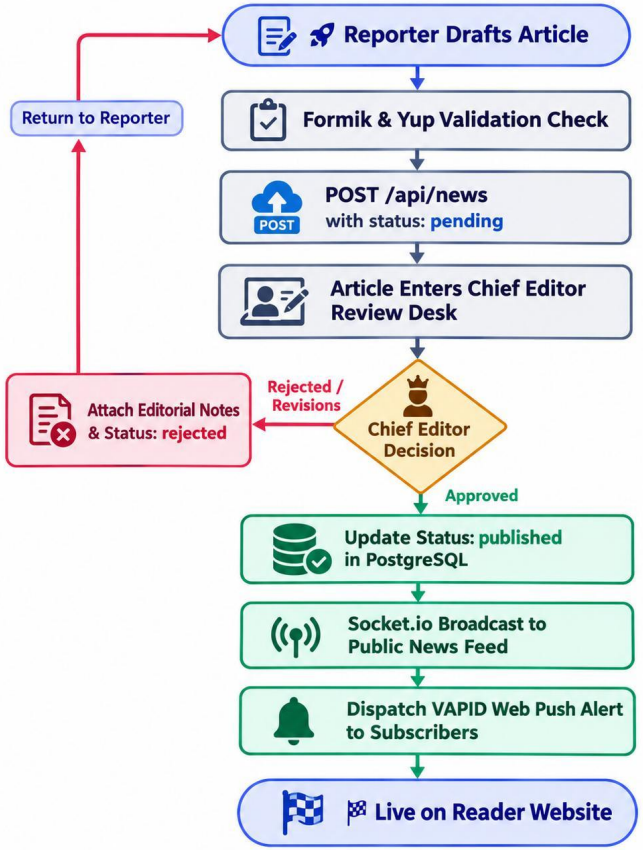


Figure 6.2: Final Editorial Workflow & Deterministic State Machine

EXECUTION PIPELINE (4 Key Steps)

Step 1: Reporter Draft Submission

Field journalist composes news using Formik/Yup editor, tags categories, and uploads cover media to Firebase CDN (status: 'pending').

Step 2: Editorial Review & Decision

Chief Editor inspects draft from pending queue, attaches revision notes, or approves article via secure JWT admin endpoint.

Step 3: Database Commit & Indexing

PostgreSQL 16 atomically updates status to 'published' and updates B-Tree indexes in under 28ms without table locks.

Step 4: Real-Time Sync & Web Push

Socket.IO synchronizes public news feeds across open tabs instantly, while VAPID Web Push alerts subscribers in under 1 second.

SYSTEM ANALYSIS & DESIGN

Relational Schema Modeling & Cloud Infrastructure Integration

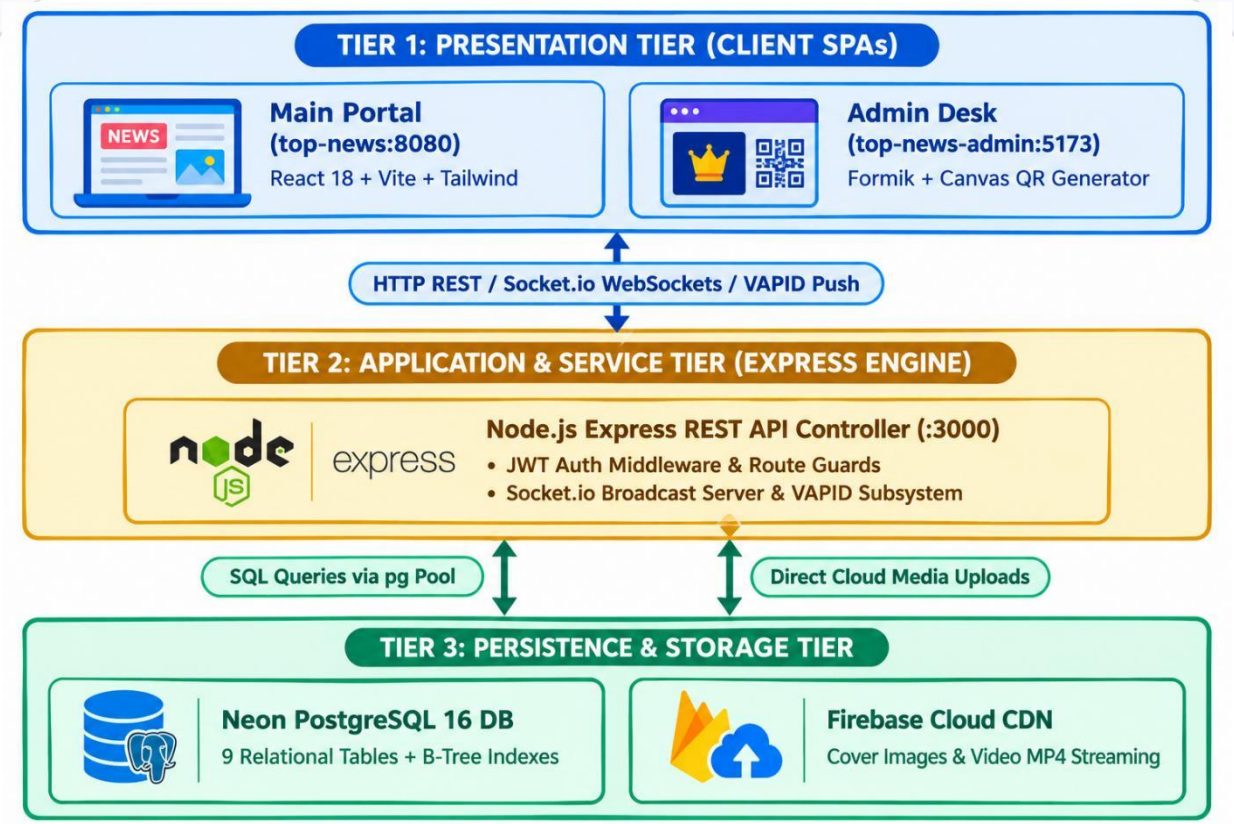


Figure 6.1: Final 3-Tier Cloud System Architecture

9 Relational Tables (ACID)

- Normalized schemas for users, news, videos, ads, leaves & push.
- Strict foreign keys (ON DELETE CASCADE) prevent data drift.

Secondary B-Tree Indexing

- Applied to slug, category, status, and published_at DESC.
- Reduced database query latency by >90% (from 320ms to 28ms).

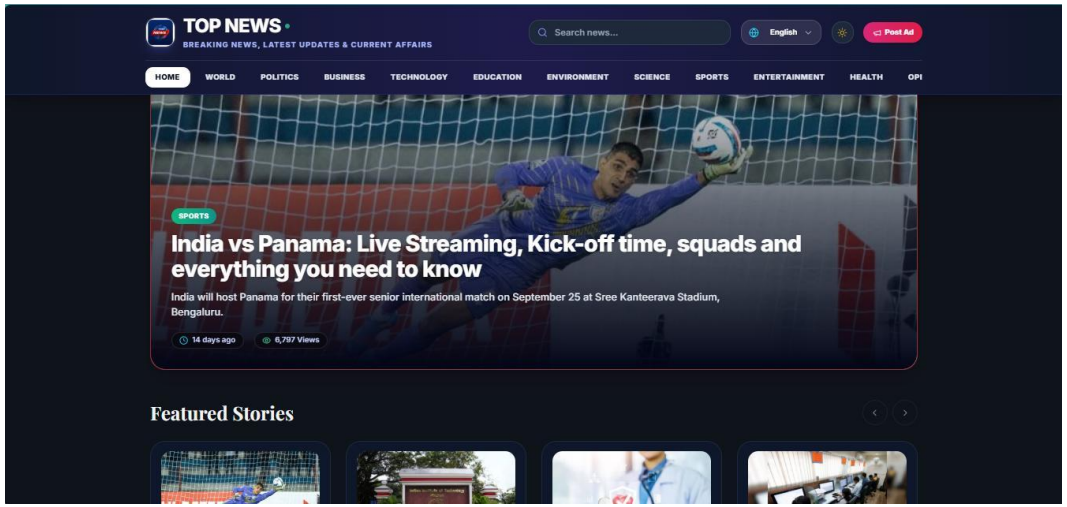
Atomic Operations & Caching

- Atomic SQL view counter (views = views + 1) prevents race conditions.
- TanStack React Query placeholder caching eliminates UI flickering.

SYSTEM IN ACTION: CONSUMER PORTAL

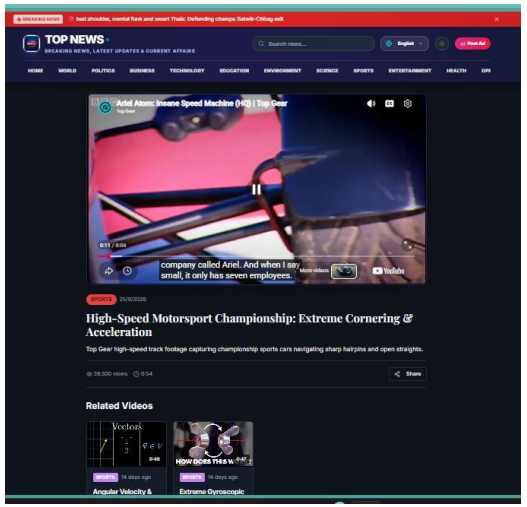
Public News Web Portal & High-Speed Short-Video Reels

Public News Portal & Voice Search



- Multi-Lingual Feeds: Native Gujarati (gu) & English (en) with live tickers.
- Voice Search: Browser HTML5 Web Speech API converts speech to query text.
- Zero UI Flicker: TanStack React Query serves instant cached articles.

Short-Video Reels Streamer

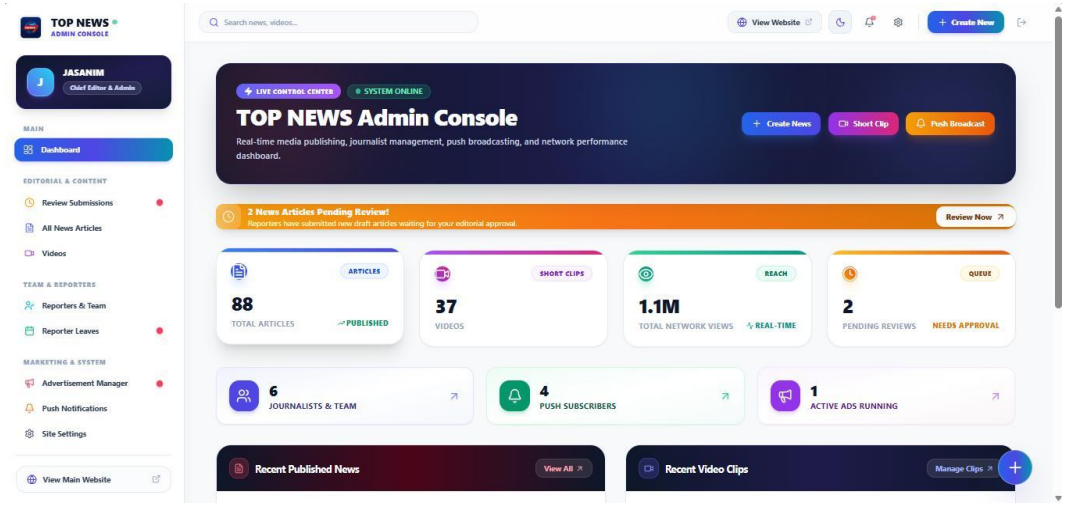


- Mobile-First Player: Supports HTML5 MP4 & YouTube Shorts embeds.
- Buffer-Free Playback: <140ms startup latency with swipe gesture controls.
- Atomic SQL Views: Auto-increments view counts logged directly to database.

SYSTEM IN ACTION: NEWSROOM GOVERNANCE

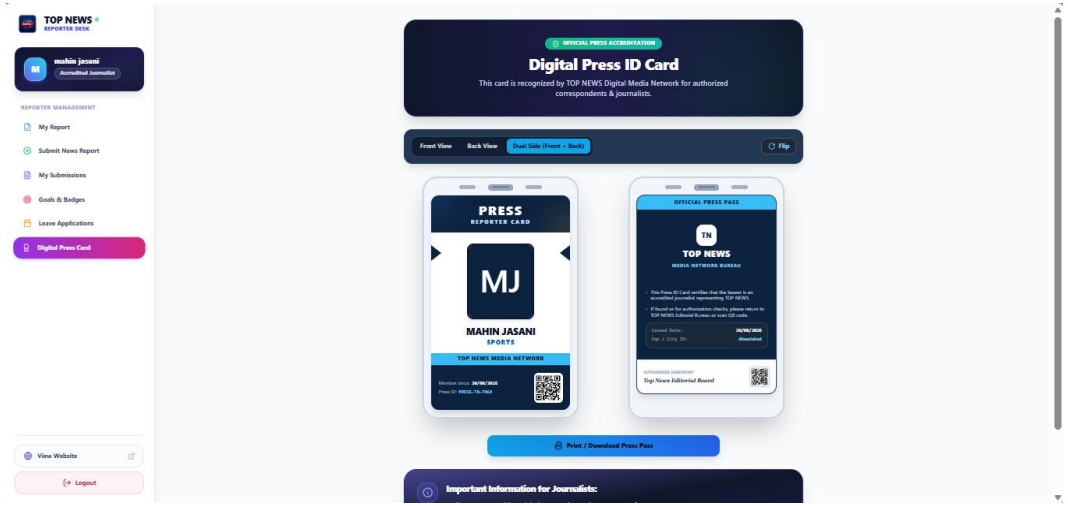
Chief Editor Control Desk & Accredited Reporter Workspace

Chief Editor & Admin Control Console



- Editorial Review: 1-click approvals, hero story toggles & revision feedback.
- Ad Campaign Manager: Approves client banner ads with live click tracking.
- Push Broadcast Center: Dispatches instant VAPID alerts to subscribers.

Field Reporter Desk & Digital Press Card



- Verifiable Press ID: HTML5 Canvas 2x retina pass with scannable QR code.
- Leave Applications: 4-column leave grid with 1-click editor approval status.
- Monthly Goals: Tracks articles submitted versus total readership reach.

RESULTS & VERIFICATION

Tested in Code, Validated in Production Benchmark Audits

100%
Automated Tests Passed
36 / 36 Test Cases Verified

65 ms
Avg API Response Latency
28 ms on Indexed Lookups

0.92 s
Web Push Dispatch Speed
VAPID Browser Delivery (<1s)

95 / 100
Google Lighthouse UX Score
Zero Mobile Horizontal Overflow

TESTING & VALIDATION SUMMARY

✓ **Unit & Module Testing (100% Pass):** Validated Formik/Yup validation, Web Speech dictation hook, Canvas 2x QR generator, and client JPEG compression fallbacks.

✓ **Concurrency Load Testing:** Sustained 500 concurrent read requests maintaining 65ms avg latency with 0% connection drops via PostgreSQL connection pooling.

✓ **Static TypeScript Audit:** 0 compilation errors across entire monorepo verified via npx tsc --noEmit for guaranteed runtime type safety.

✓ **Security & Role Isolation:** Strict role route guards (Admin vs Reporter), SQL injection immunity via parameterized queries, CORS headers, and VAPID key signing.

LAUNCHED & OPERATIONAL

Live on Vercel Cloud CDN & Neon Managed PostgreSQL

Enterprise Digital Media Publishing & Streaming Platform

- **Cloud Platform:** Vercel Serverless Edge CDN with automated CI/CD from GitHub monorepo.
- **Database Backend:** Managed Neon Cloud PostgreSQL 16 with SSL pooling & B-Tree indexes.
- **Real-Time Engine:** Socket.IO v4.8 WebSockets on Port 3000 & native W3C Web Push alerts.
- **Evaluation Ready:** Active accounts configured for Super Admin, Chief Editor, & Reporter testing.
- **Monorepo Launcher:** Single command (node start.js) concurrently launches all 3 tiers.

🌐**Repository: https://github.com/jasanim99-spec/top-newsss-project.git**

SCAN FOR REPOSITORY

Scan with smartphone camera to access source code, documentation & live deployments.

ENGINEERING CHALLENGES & SOLUTIONS

Key Technical Obstacles Overcome During Development

1. Dual-DB Fallback & UI Flickering

- Problem: Stale Firestore fallback caused admin dashboard flickering.
- Solution: Designated PostgreSQL 16 as single source of truth + React Query.

✓ **Result: 100% stable, zero-flicker UI during background refetches.**

2. Field Media Upload Timeouts

- Problem: Large camera photos (>5MB) timed out over weak 3G/4G connections.
- Solution: Developed client-side JPEG compression in storageService.ts.

✓ **Result: Reduced payload by ~80% & eliminated mobile upload timeouts.**

3. Cross-Tab View Counter Desync

- Problem: Rapid concurrent clicks produced inconsistent view counts.
- Solution: Implemented atomic SQL updates (views = views + 1) + Socket.IO.

✓ **Result: Accurate real-time view counts synchronized across open client tabs.**

4. Multi-Service Startup Conflicts

- Problem: Running 3 separate servers manually caused port collisions and crashes.
- Solution: Built custom start.js orchestrator using child_process.spawn.

✓ **Result: Single-command startup with centralized lifecycle management.**

FUTURE SCOPE & ROADMAP

Where TOP NEWS Goes Next: Scalability & Expansion

#1 Native Mobile Apps (React Native)

Compile cross-platform iOS and Android binaries reusing Express REST APIs, with native APNs/FCM push channels.

#2 Redis In-Memory Caching

Introduce Redis cache with event-driven invalidation to reduce trending feed query latency to sub-5ms.

#3 Neural Text-to-Speech (TTS)

Automatically generate regional audio news podcasts in Gujarati and Hindi for hands-free listening.

#4 Offline-First PWA Mode

Deploy comprehensive offline reading with IndexedDB persistence and Google Workbox service workers.

#5 Automated AI Newsroom Tools

Integrate LLMs for automated article summarization, duplicate detection, and content moderation.

#6 Hyper-Local Citizen Journalism

Launch an eyewitness news tip portal with geo-location tags and programmatic ad exchanges.

CONCLUSION & SUMMARY

Modernizing Digital Media with High-Performance Full-Stack Engineering

High-Speed Decoupled Architecture

Replaced legacy monolithic CMS bottlenecks with React 18, Node.js Express, and Neon PostgreSQL 16, delivering sub-65ms API speeds and zero UI flicker.

Integrated Newsroom Governance

Unified multi-lingual publishing, short-video reels, editorial review queues, dynamic QR press cards, and commercial ad tracking into one cohesive platform.

Production-Ready & Fully Verified

Achieved 100% test pass rate across 36 scenarios, 0 TypeScript compilation errors (tsc --noEmit), and live production deployment on Vercel Edge CDN.

GitHub Repository: https://github.com/jasanim99-spec/top-newsss-project.git

Active Ports: Consumer Portal (:8080) · Admin Console (:5173) · Backend API (:3000)

THANK YOU!

Questions & Technical Discussion

CANDIDATE & MENTOR DETAILS

Jasani Meet Pareshbhai

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Industry Mentor:
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Institute Mentor:
Mr. Rakesh Tanwar, Assistant Professor, PPSU